

VANSH SINGH

✉ vanshsingh@umass.edu — 🌐 github.com/vsingh2005 — Chicago, IL / Amherst, MA

Education

University of Massachusetts Amherst, Riccio College of Engineering

Bachelor of Science in Computer Engineering

Amherst, MA

Graduation: May 2026

University of Massachusetts Amherst, Isenberg School of Management

Master of Science in Business Analytics

Amherst, MA

Graduation: May 2027

Technical Skills

Programming & Systems: Python, JavaScript (ES6+), SQL, Java, Bash, SystemVerilog, Qiskit

Embedded & Hardware: Embedded C, ARM Cortex, ESP32/Arduino, I2C/SPI/UART, PCB Simulation (LTSpice)

Cloud & Infrastructure: Amazon Web Services (AWS EC2, S3, Lambda, IAM, Braket), Google Cloud Platform (GCP), Docker, CI/CD, Linux/Unix, Git

AI, ML & Analytics: PyTorch, PennyLane, Neural Networks, High-Volume Data Processing, Data Modeling, REST APIs

Professional Experience

University of Massachusetts Amherst IT

Cloud Infrastructure Engineering Intern

Amherst, MA

September 2025 – Present

- Architect and deploy scalable cloud infrastructure and automation pipelines across AWS and hybrid environments using Terraform and CI/CD workflows, reducing provisioning time by **40%** and eliminating manual configuration drift.
- Develop automated telemetry, monitoring, and health-check services in Python and Bash, improving service uptime and fault isolation across multi-tenant academic and administrative IT platforms.
- Engineer reusable Infrastructure-as-Code (IaC) modules and containerized microservices (Docker) to standardize deployment protocols across 6 cross-functional engineering teams.
- Collaborate in an Agile/Scrum environment, contributing to sprint planning, architectural design reviews, and automated integration testing.

Stem Studio

Systems & Embedded Engineering Lead / Instructor

Chicago, IL

May 2023 – Present

- Spearheaded emergency incident response and security remediation for a server-side WordPress cloaking and backdoor exploit on production infrastructure, successfully restoring environment stability and resolving critical WSOD dependencies.
- Designed and implemented hands-on embedded computing and IoT curricula (microcontrollers, circuit design, sensor interfacing, and robotics) for 150+ students, increasing project pass rates by **35%**.
- Authored and standardized technical documentation, hardware lab guides, and code repositories to ensure reproducible instruction.

CodeDay

Technical Mentor & Lead Instructor

Global Initiative

August 2020 – August 2023

- Mentored 20+ student engineering teams through end-to-end software development lifecycles, covering data structures, API integrations, algorithms, and Git version control workflows for AI and web applications.

Engineering Projects & Leadership

ASME UMass Mechatronics Team | *SolidWorks, CNC Machining, Embedded Telemetry, 3D Printing*

2nd Place Global

- Led the mechanical and electrical prototyping subteam in designing a remote payload delivery drone and articulated off-terrain rover for the ASME IAM3D Global Competition.
- Engineered custom CNC-machined lightweight structural components and integrated embedded motor drivers and telemetry feedback loops, outperforming 40+ international universities to win **2nd Place Globally**.

Quantum-Classical Hybrid Neural Network | *PyTorch, PennyLane, AWS Braket, Python*

November 2025 – Present

- Architected an end-to-end hybrid neural network coupling classical convolutional layers with parameterized quantum variational circuits, implementing Linear Combinations of Unitaries (LCU) for advanced quantum feature extraction.
- Engineered distributed training pipelines on AWS Braket, achieving **28% faster epoch convergence** and superior parameter efficiency compared to classical baseline networks.

U.S. Department of Energy – Offshore Wind Turbine System | *LTSpice, Power Electronics*

September 2025 – Present

- Led a 6-engineer electrical subteam in designing and simulating a 24V power management, regulation, and sensor telemetry system for the U.S. DOE Collegiate Wind Competition.
- Conducted circuit simulations in LTSpice and optimized composite fabrication workflows, achieving **100% compliance** with DOE electrical safety and load capacity benchmarks.